Package 'rfcipDemand'

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Type Package

Title Estimate Federal Crop Insurance Program Demand Models

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Author Francis Tsiboe [aut, cre] (<https://orcid.org/0000-0001-5984-1072>), TERRA ANALYTICS LLC [cph, fnd]

Maintainer Francis Tsiboe <ftsiboe@hotmail.com>

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Description Tools to construct county–crop–practice–plan–unit panels from the USDA RMA Summary of Business (SOBTPU) and related sources, and to estimate FCIP demand systems with two-way cluster-robust covariance. The pipeline standardizes coverage measures, merges price and instrument variables, adds rental-rate and price-index controls, reconciles county acreage (FSA/NASS), and produces diagnostics including robust first-stage F-tests. Methods align with the empirical design in ``The crop insurance demand response to premium subsidies Evidence from U.S. Agriculture" (Food Policy, 2023, 119(3)).

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URL https://github.com/ftsiboe/rfcipDemand

BugReports https://github.com/ftsiboe/rfcipDemand/issues

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`adjust_agent_outcomes_by_elasticity` *Adjust FCIP outcomes by price elasticities*

Description

Simulates changes in Federal Crop Insurance Program (FCIP) outcomes- coverage level, insured acres, liability, premium, subsidy, and indemnity- under an alternative premium-per-liability using elasticities of insured acres and/or coverage level.

Usage

```
adjust_agent_outcomes_by_elasticity(
  alternate_premium_per_liability,
  insured_acres_elasticity,
  coverage_level_elasticity,
  baseline_coverage_type,
  baseline_coverage_level,
  baseline_insured_acres,
  baseline_liability_per_acre,
  baseline_premium_per_liability,
  baseline_subsidy_per_premium,
  baseline_indemnity_per_acre,
```

```
  baseline_subsidy_percent,
  baseline_rate_differential,
  final_revenue_per_acre,
  assumption = 0,
  premium_subsidy_schedule = NULL,
  rate_differential_schedule = NULL
)
```

Arguments

`alternate_premium_per_liability`	Numeric. Alternative premium per dollar of liability.
`insured_acres_elasticity`	Numeric. Elasticity of insured acres w.r.t. price (percent basis).
`coverage_level_elasticity`	Numeric. Elasticity of coverage level w.r.t. price (percent basis).
`baseline_coverage_type`	Character. Baseline coverage type code (`"A"` or `"C"`).
`baseline_coverage_level`	Numeric in (0,1]. Baseline coverage level share.
`baseline_insured_acres`	Numeric. Baseline insured acres.
`baseline_liability_per_acre`	Numeric. Baseline liability per acre.
`baseline_premium_per_liability`	Numeric. Baseline premium per dollar of liability.
`baseline_subsidy_per_premium`	Numeric in (0,1). Subsidy share of total premium.
`baseline_indemnity_per_acre`	Numeric. Baseline indemnity per acre.
`baseline_subsidy_percent`	Numeric. Baseline subsidy percent.
`baseline_rate_differential`	Numeric. Baseline rate differential factor.
`final_revenue_per_acre`	Numeric. Revenue per acre (used for indemnity adjustment).
`assumption`	Integer (0,1,2,3). Scenario selector (see above). Default `0`.
`premium_subsidy_schedule`	Optional numeric vector of length 8 corresponding to coverage levels `0.50`, `0.55`, ..., `0.85`. Multiplicative factors applied to `baseline_subsidy_per_premium` at the scenario coverage. Defaults to 1's.
`rate_differential_schedule`	Optional numeric vector of length 8 corresponding to coverage levels `0.50`, `0.55`, ..., `0.85`. Multiplicative factors applied to `alternate_premium_per_liability` at the scenario coverage. Defaults to 1's.

Details

Assumptions (set via `assumption`):

- `0`: Fixed demand - both coverage and acres remain at baseline.

- 1: Acres respond to price (elasticity gamma); coverage fixed.
- 2: Coverage responds to price (elasticity tetha); acres fixed.
- 3: Both acres and coverage respond.

Price response:

Price response is applied using the percent change in price:

$$\%\Delta p = 100 \times \left(\frac{\text{alt}}{\text{base}} - 1 \right),$$

and for any baseline quantity q with elasticity e_q:

$$q_{\text{new}} = q_{\text{base}} \left[1 + \left(e_q \times \frac{\%\Delta p}{100} \right) \right].$$

Coverage adjustment:

Coverage is rounded to the nearest 0.05 grid and truncated to the range $(0, 0.85)$. Values less than 0.50 are set to `0` (i.e., no coverage). Coverage type code is set to `"A"` for buy-up coverage and `"C"` for CAT coverage. For CAT policies, scenario coverage levels at 0.50/0.55 are adjusted to reflect CAT rules.

Per-acre liability and indemnity are updated using `adjust_indemnity_liability_per_acre`.

Optional schedules:

Coverage-specific scaling is possible via:

- `premium_subsidy_schedule`: length-8 numeric for coverage levels `0.50`, `0.55`, ..., `0.85`. Multiplies the `baseline_subsidy_per_premium` at the scenario coverage.
- `rate_differential_schedule`: length-8 numeric for the same grid. Multiplies the `alternate_premium_per_li` at the scenario coverage.

Missing or zero inputs:

If either `insured_acres` or `coverage_level_percent` is `NA` or `0` after adjustment, both are reset to `0` and all dollar outcomes are set to `0`.

Elasticities are applied multiplicatively to baseline quantities. Coverage levels are snapped to the 0.05 grid and truncated. When schedules are supplied, subsidy shares and premium rates are rescaled relative to the baseline coverage before applying to the scenario coverage.

Value

A list with elements:

`coverage_type_code` Scenario coverage type (`"A"` or `"C"`).

`coverage_level_percent` Scenario coverage level share.

`insured_acres` Scenario insured acres.

`liability_amount` Total liability = acres * adjusted liability per acre.

`total_premium_amount` Total premium = liability * alt premium/liability.

`subsidy_amount` Subsidy = total premium * subsidy share.

`indemnity_amount` Total indemnity = acres * adjusted indemnity per acre.

`price_change_pct` Percent price change used for elasticity adjustments.

`adjust_indemnity_liability_per_acre` *Adjust liability and indemnity per acre at an alternative coverage level*

Description

Compute adjusted liability per acre and adjusted indemnity per acre when moving from a baseline coverage level to a new level. The method floors production-to-count at 0 and resolves five mutually exclusive cases via `data.table::fcase()`.

Usage

```
adjust_indemnity_liability_per_acre(
  coverage_level_percent,
  final_revenue_per_acre,
  baseline_coverage_level,
  baseline_liability_per_acre,
  baseline_indemnity_per_acre
)
```

Arguments

`coverage_level_percent`
: Numeric vector; alternative coverage level (proportion in (0,1)).

`final_revenue_per_acre`
: Numeric vector; revenue per acre at harvest (used when baseline had no indemnity and baseline coverage is lower).

`baseline_coverage_level`
: Numeric vector; baseline coverage level (proportion in (0,1)).

`baseline_liability_per_acre`
: Numeric vector; baseline liability per acre.

`baseline_indemnity_per_acre`
: Numeric vector; baseline indemnity per acre.

Details

- `production_to_count = pmax(baseline_liability_per_acre - baseline_indemnity_per_acre, 0)`
- `adj_Liability_per_acre = (baseline_liability_per_acre / baseline_coverage_level) * coverage_level_percent`
- Indemnity cases (I1-I5) selected with `data.table::fcase()`.
- Adjusted indemnity is floored at 0 in the two cases that can go negative (coverage falls with a baseline loss; coverage rises with no baseline loss).

Value

A list with:

`adj_Liability_per_acre` Adjusted liability per acre.

`adj_Indemnity_per_acre` Adjusted indemnity per acre.

`calibrate_fcip_demand_elasticities` *Calibrate FCIP demand elasticities*

Description

Fits a 2-equation FCIP demand system over an estimation window ending in `calibration_year` and returns capped elasticities (-2, 0) by disaggregation level using `fcip_demand_sys_estimate()`

Usage

```
calibrate_fcip_demand_elasticities(
  calibration_year,
  estimation_window,
  data,
  drawn_pools = NULL,
  disaggregate = NULL,
  control = rfcipDemand_controls()
)
```

Arguments

`calibration_year`	Integer. Last year of the estimation window.
`estimation_window`	Integer (>= 1). Number of years ending at `calibration_year`.
`data`	`data.table`. Input panel. Must include: `commodity_year`, `commodity_code`, `drawID`, `pool`, `insurance_plan_code`, `price`, `net_reporting_level_amount`, `coverage_level_percent_aggregate`, `rent`, `county_acreage`, `total_premium_amount`, `subsidy_amount`, `liability_amount`, `tau`, `subsidy_rate_65`, `subsidy_rate_75`.
`drawn_pools`	Optional `data.table` used to filter `data` via an inner join on intersecting columns.
`disaggregate`	Optional character vector of grouping variables passed to the estimator (e.g., `"commodity_code"` or `c("state","commodity_code")`).
`control`	a list of control parameters. see `rfcipDemand_controls`.

Details

Internally, the function (i) restricts `data` to the estimation years, (ii) collapses insurance plan codes and sets `price := 1` for non-RP crops, (iii) constructs log variables, trend and year dummies, and (iv) estimates the system with `rate` endogenous and `tau0` as the excluded instrument. Estimates are truncated to (-2, 0) and returned in wide form.

Value

`data.table` with columns: `disag`, `level`, `gamma_elasticity`, `theta_elasticity`.

`estimate_fcip_instruments` *Estimate FCIP Instrumental Variables (Unloaded Rates)*

Description

Uses historical FCIP rate data to build instrumented unloaded-rate variables following:

1. Tsiboe & Turner (2023), Econometric identification of crop insurance participation *Agricultural and Resource Economics Review*, 52(3):476-497. https://doi.org/10.1017/age.2023.13

Usage

```
estimate_fcip_instruments(year, statplan)
```

Arguments

`year`	Integer. The target crop year for which to construct instruments.
`statplan`	A data.table containing FCIP rate elements, including at least: **commodity_year** Year of the rate observation. **state_code, county_code** County identifiers. **commodity_code** Crop identifier. **insured_area, lcr, contiguous_state_code, contiguous_county_code** Fields required by `estimate_fcip_unloaded_rate()`.

Details

1. **Task list**: Identify all unique (state, county) pairs with data in the 2-21 years before `year`.
2. **Unloaded-rate calculation**: For each county in `task_list`, call `estimate_fcip_unloaded_rate()` on the same 2-21 year window to get `tau`. Errors return `NULL` so processing continues.
3. **Contiguous-county smoothing**:
   - Build a lookup table of contiguous counties (using `contiguous_county`).
   - For each contiguous group, compute the mean `tau` to get `tau_c`.
4. **Merge & fill**: Left-join the raw `adm` and `contiguous_adm`; replace any zero/NA/Inf `tau` with the group mean `tau_c` into `tau_sob`.
5. **Cleanup**: Drop helper columns (`tau`, `tau_c`), remove invalid rows, add `commodity_year`, and return the result.

Value

A data.table with one row per county-crop for the specified `year`, containing:

state_code, county_code, commodity_code Keys.

tau_sob Smoothed unloaded rate (uses contiguous-county means to fill zeros/NAs).

commodity_year The input `year`, repeated.

See Also

Other FCIP instruments: `get_yu2018_instrument()`

fcip_contiguous_county

Contiguous counties

Description

County adjacency from the RMA Actuarial Data Master, used to build neighbouring-county instruments in `estimate_fcip_instruments()`.

Usage

```
fcip_contiguous_county
```

Format

A data frame with one row per county and contiguous county:

record_type_code, record_category_code ADM record identifiers.

reinsurance_year Reinsurance year of the ADM release.

state_code, county_code FIPS codes of the county.

contiguous_state_code, contiguous_county_code FIPS codes of the contiguous county.

contiguous_county_type_code ADM contiguity type code.

last_released_date, released_date, deleted_date ADM record dates.

data_source Source description.

Source

USDA Risk Management Agency, Actuarial Data Master, record A01230.

fcip_demand_data_dispatcher

Build dataset to estimate Federal Crop Insurance Program (FCIP) demand (modular pipeline)

Description

End-to-end pipeline that: (1) prepares SOBTPU and coverage aggregates, (2) adds prices/instruments/rental rates/price index, (3) reconciles county acreage (FSA + NASS), and (4) finalizes bins/labels/pooling for demand estimation.

Usage

```
fcip_demand_data_dispatcher(
  study_years = 2001:(as.numeric(format(Sys.Date(), "%Y")) - 1),
  identifiers = c("commodity_year", FCIP_INSURANCE_POOL, "insurance_plan_code",
    "unit_structure_code")
)
```

Arguments

study_years	Integer vector of commodity years to include. Defaults to 2001:(as.numeric(format(Sys.Date(), "%Y")) - 1).
identifiers	Character vector of grouping keys that define the aggregation grain. Must be columns present in SOBTPU (e.g., "commodity_year", FCIP_INSURANCE_POOL, "insurance_plan_code", "unit_structure_code", and-if desired-additional keys like "commodity_code" or "practice_code"). Enrichment joins for re-codes are performed only when the required keys are included in identifiers

Details

Aligned with Asche, Bekkerman, & Li (2023), *Food Policy*, 119(3):102505 (doi:10.1016/j.foodpol.2023.102505). Requires internet access to download release .rds assets and several in-memory lookup tables (see stage docs).

Value

A data.table ready for FCIP demand estimation.

---

fcip_demand_elasticities_lavaan

Calibrate FCIP demand elasticities via IV-SEM (lavaan)

Description

Fits a just-identified IV-style SEM where instr_rate instruments tilda_rate, and tilda_rate enters two outcome equations (tilda_ghamma, tilda_theta). Endogeneity is encoded via residual correlations between tilda_rate and each outcome. The instrument is excluded from the outcome disturbances.

Usage

```
fcip_demand_elasticities_lavaan(
  data,
  estimator = c("ML", "MLR"),
  missing = c("fiml", "listwise"),
  control = rfcipDemand_controls()
)
```

Arguments

data	data.frame with cols: instr_rate, tilda_rate, tilda_gamma, tilda_theta
estimator	"ML" or "MLR" (default "MLR" = robust SE/tests)
missing	"fiml" or "listwise" (default "fiml")
control	a list of control parameters. see rfcipDemand_controls.

Details

Constraints imposed:

1. b1 < 0, 2) b2 < 0, 3) b1 + b2 + b1*b2 < 0

Value

A data.table of parameter estimates and logical convergence flag

---

fcip_demand_sys_coeff_table

Tidy coefficient table with cluster-robust SEs (from supplied VCOV)

Description

Builds a clean coefficient table for a `systemfit` model using a **user-supplied covariance matrix** (e.g., two-way clustered from `fcip_demand_sys_vcov()`). Estimates come from `coef(fit)`, standard errors from `diag(vcMat)`, then Z-scores and two-sided normal p-values are computed. The `demand` column is inferred from the equation prefix in the coefficient names:

- `"gamma_*"` to `"gamma"`
- `"theta_*"` to `"theta"` Otherwise the prefix itself is used.

Usage

```
fcip_demand_sys_coeff_table(fit, vcMat, p_digits = 5)
```

Arguments

`fit`	A fitted `systemfit` object.
`vcMat`	A covariance matrix conformable with `coef(fit)`. Row/column names are used to align; if missing, positional alignment is assumed.
`p_digits`	Integer; number of digits to keep for p-values (default `5`).

Value

A `data.frame` with columns: `demand`, `coef`, `Estimate`, `StdError`, `Zvalue`, `Pvalue`.

---

fcip_demand_sys_effect

Delta-method "total protection response"

Description

Combines equation-specific effects into a single "total" effect for each regressor in `c(fields$endogenous, fields$included)` using `car::deltaMethod` and a supplied covariance matrix.

Usage

```
fcip_demand_sys_effect(fit, vcMat, fields, data)
```

Arguments

fit	A fitted systemfit object (the structural system).
vcMat	Covariance matrix conformable with coef(fit) (e.g., from fcip_demand_sys_vcov()).
fields	Named list carrying model fields; must include outcome, endogenous, and included.
data	Estimation data used to check variable availability and build delta-method expressions.

Value

A data.frame with rows demand="total" and columns: demand, coef, Estimate, StdError, Zvalue, Pvalue.

---

fcip_demand_sys_estimate

System estimator (modular wrapper; preserves original outputs)

Description

Estimates a two-equation system with an endogenous regressor across disaggregation levels. The pipeline is: (1) per-level sample selection (min n per commodity_year), (2) partialling-out / tilda transforms, (3) system estimation (e.g., IV/3SLS via internal helpers), (4) clustered variance estimation, (5) delta-method totals, (6) optional constrained re-estimation of elasticities (lavaan), (7) diagnostics, and (8) row-binding results across levels.

Usage

```
fcip_demand_sys_estimate(
  model,
  data,
  constrained_elasticities = FALSE,
  control = rfcipDemand_controls()
)
```

Arguments

model	List specifying the system; required elements are: • outcome (character(2)): names of the two outcomes in data. • endogenous (character(1)): endogenous regressor name. • included (character): included (exogenous) regressors. • disag (character(1) or NULL): disaggregation key column in data. • FE (logical): include fixed effects in the internal run (handled by helpers). Optional elements: • excluded (character or NULL): excluded instruments. • partial (character or NULL): variables to partial out (tilda). • restrict (logical): pass-through to internal restricted estimation. • name (character(1)): label carried to the output.

`data` `data.frame`/`data.table` containing all referenced columns in `model` plus `pool` and `commodity_year`. Columns in `model$outcome`, `model$endogenous`, `model$included`, and (if used) `model$excluded` and `model$partial` must exist in `data`.

`constrained_elasticities`
Logical (default `FALSE`). If `TRUE`, re-estimate the elasticities via a constrained SEM (lavaan) and, where applicable, replace positive elasticity estimates with constrained ones. See **Constrained elasticities (optional)**.

`control` a list of control parameters. see `rfcipDemand_controls`.

Details

Inputs and preprocessing

- `model$outcome` must be a character vector of length 2 giving the two outcome column names in `data`. Internally these are mapped to `gamma` and `theta` for estimation convenience.
- `model$disag` is the name of the disaggregation key. If `NULL`, a dummy `"full_sample"` key is created.
- The disaggregation key is coerced to `character`.
- For each level of `model$disag`, the function keeps only levels that have **at least 30 observations per** `commodity_year`. Levels failing this threshold are dropped.

Per-level estimation For each retained level, the function calls internal helpers (`fcip_demand_sys_run`, etc.) to (i) partially out controls if requested, and (ii) estimate the system with clustered VCOV and delta-method totals. Errors at the level are caught and the level is skipped (no hard stop).

Constrained elasticities (optional) When `constrained = TRUE`, elasticities on the endogenous regressor are re-estimated per level via a lavaan SEM with sign restrictions (internal helper `fcip_demand_elasticities_l` using `estimator = "MLR"`, `missing = "listwise"`). The constrained estimates replace any positive system estimates for the elasticities; a logical flag `constrained` marks levels where a replacement occurred. The returned columns are reduced to `disag`, `level`, `demand`, `constrained`, and `Estimate`.

Returned shape

- If `constrained_elasticities = FALSE` (default), returns the full per-level system output from `fcip_demand_sys_run` with additional columns: `disag`, `level`, and rounded `Zvalue`, `Pvalue`.
- If `constrained = TRUE`, returns a compact table with `disag`, `level`, `demand`, `constrained`, `Estimate` after merging the constrained elasticities.

Value

A `data.table` aggregating results across all disaggregation levels. The column set depends on `constrained` (see **Returned shape**).

fcip_demand_sys_fit *Build systemfit formulas and estimate the system*

Description

Constructs the list of structural equations (g) and instrument sets (h), then runs `systemfit()` using OLS (when no excluded instruments) or 3SLS-GMM (when excluded instruments are present).

Usage

```
fcip_demand_sys_fit(
  data,
  fields,
  tilda_included,
  tilda_endogenous,
  tilda_excluded
)
```

Arguments

`data`	Estimation `data.frame`/`data.table` containing the `tilda_*` and `instr_*` variables referenced by the formulas.
`fields`	Named list with at least `outcome`, `included`, `endogenous`, and optionally `excluded`.
`tilda_included`	Character vector of residualized included regressor names (e.g., `"tilda_x1"`).
`tilda_endogenous`	Character vector of residualized endogenous regressor names (e.g., `"tilda_z1"`).
`tilda_excluded`	Character vector of instrument names (e.g., `"instr_z1"`), or `NULL` when no excluded instruments are used.

Value

A list with elements:

`fit`	Fitted `systemfit` object.
`g`	List of structural formulas.
`h`	List of instrument formulas.

fcip_demand_sys_partial *Residualize ("partial out") and build tilded / instrument variables*

Description

If excluded instruments exist, runs first-stage OLS for each endogenous variable e: `e ~ 1 + partial + included + excluded`, storing the fitted values as `instr_e`. If `partial` is non-empty, it then regresses `instr_e ~ 1 + partial` and replaces `instr_e <- instr_e - fitted(instr_e ~ partial)` (i.e., removes the partial component; conceptually $\widehat{\text{instr}_e}(\text{partial})$).

Outcomes, included, and endogenous variables are residualized on `partial` to create `tilda_<var>` (or copied if `partial` is empty).

Usage

```
fcip_demand_sys_partial(data, fields, partial_override = NULL)
```

Arguments

data	A data.frame/data.table with referenced variables.
fields	List with: outcome, endogenous, included, optional excluded, optional partial.
partial_override	Optional character vector to override fields$partial.

Details

Uses defensive checks so absent columns are ignored (with a warning) rather than erroring. If partial is empty, residualization is a no-op and tilda_* simply copy the originals. Formulas are constructed via stats::reformulate() to avoid paste/quoting pitfalls.

Value

List with data, tilda_included, tilda_endogenous, tilda_excluded.

---

fcip_demand_sys_prep *Prepare data for analysis*

Description

Drops incomplete/invalid rows, removes constant partials, and optionally demeans via a fixed-effects helper.

Usage

```
fcip_demand_sys_prep(data, fields)
```

Arguments

data	Estimation dataset that already contains all columns referenced by fields.
fields	Named list: outcome, endogenous, included, excluded (opt), partial (opt), FE (logical), disag (column name).

Value

A list: data (prepped), NFE (number of FE), partial (possibly reduced).

fcip_demand_sys_run *Orchestrate analysis*

Description

Runs the full pipeline: prep -> partial/tilda creation -> systemfit -> two-way clustered VCOV -> delta-method totals -> optional restricted step -> diagnostics; then returns a tidy coefficient table with metadata.

Usage

```
fcip_demand_sys_run(data, fields)
```

Arguments

data	Estimation dataset
fields	Named list carrying model fields (see fcip_demand_estimation()), including disag, FE, outcome, endogenous, included, optional excluded, partial, restrict, and name.

Value

A data.frame with columns demand, coef, Estimate, StdError, Zvalue, Pvalue and meta-columns model, endogenous, FE, name, disag

fcip_demand_sys_tests *System diagnostics: two-way robust first-stage F (+ optional approx. J)*

Description

Produces diagnostics **without** re-running GMM:

- **FTest**: joint relevance of excluded instruments in each first stage, using the same two-way (pool by crop year) cluster-robust covariance via fcip_demand_sys_vcov() with kind = "lm". Reports the **minimum** F across endogenous regressors.
- **JTest** (optional): an *approximate* over-identification test computed as the sum of per-equation Sargan statistics $J_k \approx n_k R_k^2$ from regressions of equation residuals on that equation's instrument set. This is a quick check (not the system Hansen J).

Usage

```
fcip_demand_sys_tests(g, h, data, fit, NFE, approx_j = FALSE)
```

Arguments

g	List of system equations (the same formulas passed to systemfit).
h	List of instrument formulas (the same formulas passed to systemfit).
data	Estimation data.frame/data.table containing all variables in g/h plus clustering columns pool and crop_yr.
fit	A fitted systemfit object (used for N and residCov_* extraction).
NFE	Integer: number of absorbed fixed effects (for reporting only).
approx_j	Logical, compute the approximate (non-robust) Sargan J as described above. Default FALSE (returns NA for JTest).

Value

A data.frame with rows: N, NFE, residCov_11, residCov_22, residCov_12, JTest, FTest.

---

fcip_demand_sys_vcov *Two-way cluster-robust covariance for FCIP demand models*

Description

Computes a Cameron-Gelbach-Miller two-way cluster-robust covariance matrix using inclusion-exclusion: $V = V_{pool} + V_{year} - V_{pool_year}$. Works for both systemfit (stacked system) and lm (first-stage).

Usage

```
fcip_demand_sys_vcov(
  object,
  data,
  kind = c("systemfit", "lm"),
  pool_col = "pool",
  year_col = "commodity_year",
  NFE = 0L,
  n_partial = 0L,
  n_eq = NULL
)
```

Arguments

object	Fitted model: either a systemfit or lm.
data	Estimation data containing pool and year identifiers.
kind	One of c("systemfit","lm"). If omitted, auto-detected.
pool_col	Name of the pool/cluster id column in data (default "pool").
year_col	Name of the year/time id column in data (default "crop_yr").
NFE	Integer; number of absorbed fixed effects (for df rescaling).
n_partial	Integer; count of variables partialed out per equation.
n_eq	Integer; number of equations (length(object$eq) for systemfit, 1 for lm). You can override if needed.

Details

Rescaling. Let n be the number of observations (stacked across equations for `systemfit`). With k_old the number of coefficients and $k_new = k_old + NFE + n_partial * n_eq$, the returned matrix is scaled by $(n - k_old - 1)/(n - k_new - 1)$.

Row alignment (lm). Rows used by `lm` are inferred from `rownames(model.matrix(object))`. If they cannot be mapped back to `data`, the first `nobs(object)` rows are used.

Value

Covariance matrix aligned with `coef(object)`.

`fcip_elasticity_limits`

Bounds on FCIP demand elasticities

Description

Program-level limits used to constrain calibrated demand elasticities. Each limit is 50% of the magnitude of the corresponding program-level elasticity. Used as the default `elasticity_limits` in `rfcipDemand_controls()`.

Usage

```
fcip_elasticity_limits
```

Format

A data frame with one row and 4 columns:

gamma Limit on the elasticity of insured acres (net reporting level).

theta Limit on the elasticity of coverage level.

total Limit on the total elasticity.

data_source Text description of how the limits were derived.

Source

Estimated by `data-raw/scripts/estimate_fcip_elasticity_limits.R`.

FCIP_INSURANCE_POOL *Insurance pool identifier fields*

Description

A character vector of column names that together define a unique insurance pool in the Federal Crop Insurance Program (FCIP).

Usage

```
FCIP_INSURANCE_POOL
```

Format

A `character` vector of field names.

Details

Insurance pools represent the most granular level of rate making within FCIP. Each pool is uniquely identified by the combination of:

- **state_code**: State FIPS code
- **county_code**: County FIPS code
- **commodity_code**: Crop commodity code
- **type_code**: Crop type (e.g., grain vs. silage)
- **practice_code**: Production practice (e.g., irrigated, organic)

Value

A `character` vector specifying the columns used to define each FCIP insurance pool.

Examples

```
## Not run:
# Default insurance pool fields
FCIP_INSURANCE_POOL

# Override to a subset of the original fields
rFarmPolicySim:::FCIP_INSURANCE_POOL <- c(
  "state_code", "county_code", "commodity_code"
)

## End(Not run)
```

fcip_recodes_commodity_groupings
Commodity groupings

Description

Commodity names and groupings by commodity year, joined onto FCIP data by `fcip_demand_data_dispatcher()`.

Usage

```
fcip_recodes_commodity_groupings
```

Format

A data frame with one row per `commodity_year` and `commodity_code`:

commodity_year Crop (commodity) year.

commodity_code RMA commodity code.

CROP Harmonized crop name.

commodity_name RMA commodity name.

commodity_group Commodity group, as `"group:subgroup"`.

annimal_flag, annual_flag, perennial_flag Indicators for animal, annual and perennial commodities (`"1"`, `"0"`, or `"-Inf"` when unknown).

selected_crops_flag 1 for commodities in the package's core crop set.

data_source Source description.

Source

USDA Risk Management Agency, Actuarial Data Master, records A00400 and A00420, supplemented with legacy ADM files.

fcip_recodes_insurance_plan
Insurance plan recodes

Description

Insurance plan names and classifications by commodity year, joined onto FCIP data by `fcip_demand_data_dispatcher`

Usage

```
fcip_recodes_insurance_plan
```

Format

A data frame with one row per `commodity_year` and `insurance_plan_code`:

commodity_year Crop (commodity) year.

insurance_plan_code RMA insurance plan code.

insurance_plan_name, insurance_plan_abbreviation RMA plan name and abbreviation.

fcip_policy_type `"Basic"`, `"Endorsement"`, or `"Basic or Endorsement"`.

outcome_protected What the plan insures, e.g. `"Yield"`, `"Revenue"`, `"Dollar"`.

triger_level Loss trigger: `"Individual"` or `"Group"`.

insurance_plan_type_rma RMA plan type.

insurance_plan_recode Harmonized plan code used across years.

data_source Source description.

Source

USDA Risk Management Agency, Actuarial Data Master, record A00460, supplemented with legacy ADM files.

`fcip_recodes_practice` *Practice recodes*

Description

Practice names and irrigation, organic and fallow recodes by commodity year, joined onto FCIP data by `fcip_demand_data_dispatcher()`.

Usage

```
fcip_recodes_practice
```

Format

A data frame with one row per `commodity_year`, `commodity_code` and `practice_code`:

commodity_year Crop (commodity) year.

commodity_code RMA commodity code.

practice_code, practice_name RMA practice code and name.

irrigation_recode `"IRR"`, `"NON-IRR"` or `"Other"`.

organic_recode Organic status of the practice.

fallow_recode `"FALLOW"` or `"CONTINUOUS CROPPING"`.

data_source Source description.

Source

USDA Risk Management Agency, Actuarial Data Master, record A00510, supplemented with legacy ADM files.

`fixed_effect_model_data_prep` *Prepare and demean data for fixed-effects models*

Description

This function

1. Filters to complete cases on the specified panel, time, weight, variables, and output
2. If `output` is NULL, creates a dummy output column filled with 1s
3. Drops any panel with only one observation
4. Computes within-panel means for the output + each variable in `varlist` (`_mean_i`)
5. Computes overall sample means for the same set of variables (`_mean`)
6. Replaces each variable in `varlist` by `value - within_panel_mean + overall_mean`

Usage

```
fixed_effect_model_data_prep(
  data,
  varlist,
  panel,
  time,
  wvar = NULL,
  output = NULL
)
```

Arguments

`data`	A data.frame or data.table containing the data.
`varlist`	Character vector of variable names to be demeaned.
`panel`	Character vector of column name(s) defining the panel identifier.
`time`	Character scalar name of the time variable.
`wvar`	Character scalar name of a variable to keep but *not* demean (optional, default NULL).
`output`	Character scalar name of an output variable whose means are computed but not altered; if NULL, a dummy column named `"output"` is created (optional, default NULL).

Value

A list with components

- **data**: a data.table containing
  - the original `panel`, `time`, `wvar`, `varlist`, and `output` columns
  - two mean columns for each of `c(output, varlist)`: `<name>_mean_i` (within-panel) and `<name>_mean` (overall)
- **NFE**: the number of panels with more than one observation

format_fcip_demand_table *Table: Crop Insurance Demand System for US Federal Crop Insurance Pools (2001/22)*

Description

Build a two-column, GitHub-safe panel table summarizing a crop insurance demand system. The table is organized into panels for coverage level (Theta), insured acres (Gamma), total protection response, a covariance matrix block, and additional statistics. Coefficients are formatted as `estimate (std. error)` with significance stars.

Usage

```
format_fcip_demand_table(df, var_labels)
```

Arguments

df	A data frame containing the results with columns: • demand (chr): panel identifier; expected values include "Theta", "Gamma", and "Total". • coef (chr): raw coefficient/row labels (e.g., "tilda_rate", "residCov_11", "N"). • Estimate (dbl): point estimates. • StdError (dbl): standard errors (may be NA for scalars). • Pvalue (dbl): p-values used to add significance stars.
var_labels	A named character vector mapping raw names to display labels,

Details

Designed for README/output knitted as `github_document`; use with `knitr::kable(..., format = "pipe")` to avoid HTML-only features.

Value

A tibble with two columns, `Variables` and `Estimates`, where panel headers have empty `Estimates` to enable bolding (if rendered in HTML) and coefficients are formatted as `"estimate*** (se)"`.

get_yu2018_instrument *Formulate & Merge National Subsidy Rate Instrument (Yu et al., 2018)*

Description

Downloads the historical Summary of Business RDS and computes national subsidy-rate instruments at specified coverage levels, following Yu et al. (2018).

Usage

```
get_yu2018_instrument(
  dt,
  delivery_systems = c("RBUP", "FBUP"),
  plan_codes = c(1:3, 90, 44, 25, 42),
  coverage_levels = c(65, 75)
)
```

Arguments

dt	sobcov
delivery_systems	Character vector. Delivery systems to include; default `c("RBUP","FBUP")`.
plan_codes	Integer vector. Insurance plan codes to include; default `c(1:3, 90, 44, 25, 42)`.
coverage_levels	Numeric vector. Percent coverage levels to keep; default `c(65, 75)`.

Value

A data.table with columns: commodity_year, subsidy_rate_65, subsidy_rate_75.

See Also

Other FCIP instruments: `estimate_fcip_instruments()`

`global_variables` *fsa_crop_linker*

Description

A combined dataset for global_variables

Usage

```
global_variables
```

Format

A data frame with 8594 rows and 8 columns covering Inf–Inf.

Source

Internal innovation

nassSurveyPriceRecivedIndex *NASS index of prices received*

Description

Annual index of prices received by farmers, used as a deflator by `fcip_demand_data_dispatcher()`.

Usage

```
nassSurveyPriceRecivedIndex
```

Format

A data frame with one row per year:

commodity_year Year.

index_for_price_recived Index of prices received.

data_source Source description.

Source

USDA National Agricultural Statistics Service, Quick Stats.

nassSurveyRentalRates *State cropland rental rates*

Description

Average cropland cash rent by state and year, joined onto FCIP data by `fcip_demand_data_dispatcher()`.

Usage

```
nassSurveyRentalRates
```

Format

A data frame with one row per state and year:

NAME State name.

state_code State FIPS code.

commodity_year Year.

rent Cash rent, dollars per acre.

data_source Source description.

Source

USDA National Agricultural Statistics Service, Quick Stats, via `get_state_rental_rates()`.

premium_subsidy_schedule *Premium subsidy schedule*

Description

Premium subsidy rates by plan, coverage level, coverage type and unit structure.

Usage

```
premium_subsidy_schedule
```

Format

A data frame with columns:

commodity_year Crop (commodity) year.
insurance_plan_code RMA insurance plan code.
coverage_level_percent Coverage level, as a proportion (0.50-0.95).
coverage_type_code RMA coverage type code ("A", "C", "L").
unit_structure_code RMA unit structure code, or "ALL".
subsidy_percent Premium subsidy rate, as a proportion.
data_source Source description.

Source

USDA Risk Management Agency, Actuarial Data Master, supplemented with legacy ADM files.

rfcipDemand_controls *Demand Model Control Parameters*

Description

Constructs a control list for demand-model routines in **rfcipDemand**, primarily governing admissible ranges for elasticity parameters used in demand adjustments and simulations.

If `elasticity_limits` is not supplied, the function defaults to `fcip_elasticity_limits`, a package-level object defining program-consistent elasticity bounds.

Usage

```
rfcipDemand_controls(elasticity_limits = NULL)
```

Arguments

`elasticity_limits`
: Optional named list specifying lower and upper bounds for elasticity parameters. If `NULL`, defaults to `fcip_elasticity_limits`.

Value

A named list of control parameters, ready to be passed to other simulation functions.

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